Anne Fayolle

CONTACT INFORMATION

Department of Mathematics
University of Utah , LCB 317.

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RESEARCH INTERESTS

Algebraic geometry and commutative algebra.

EDUCATION

University of Utah

PhD in mathematics (2022 - Expected Summer 2027)
Advisor: Karl Schwede

École Polytechnique Fédérale de Lausanne (EPFL)

MSc. Mathematics, July 2022

McGill University

BSc. First class honours in mathematics, minor in computer science, May 2020

Université de Montréal

Certificate Philosophy, with mention of excellence, May 2020

PUBLICATIONS

2. Anne Fayolle, *Centers of perfectoid purity.* Journal de l'École Polytechnique Mathématique. DOI:jep.313 2025.

1. J. Carvajal-Rojas, Anne Fayolle, *On tame ramification and centers of F-purity.* Journal of the London Mathematical Society. DOI: jlms.12993 2024.

PREPRINTS

1. Hanlin Cai, Anne Fayolle, Linquan Ma, Kevin Tucker, Karl Schwede, Ilya Smirnov, and Joe Waldron. *Perfectoid pure and signature pairs.* Preprint Link 2026.
2. Anne Fayolle, Abhay Goel, Devlin Mallory, Eamon Quinlan-Gallego, and Teppei Takamatsu. *The witt vectors package for Macaulay2.* Arxiv Link 2026.

SOFTWARE

1. The Witt Vector Package for Macaulay2, joint with Abhay Goel, Devlin Mallory, Eamon Quinlan-Gallego, and Teppei Takamatsu.

SCHOLARSHIPS AND AWARDS

2025 Simons Dissertation Fellowship
40 fellowships awarded annually to 3rd year Ph.D students in mathematics by the Simons foundation, $38,400 over 2 years.
2023 Graduate Student Travel Assistance Award, University of Utah
Funds from the department of Mathematics to support travel to a conference.
2023 NSERC CGRS-D,
Fellowship awarded by the Canadian government to the top doctoral students applying in the sciences, 105,000 CAD over 3 years.
2022 EDGES fellowship University of Utah
Funds awarded by the college of Science at the University of Utah to recruit, retain, and promote the success of outstanding graduate students, $10,000 over 5 years.
2019 FRQNT supplement to the NSERC award,
Fellowship awarded by the Quebec government to undergraduate students to pursue research for a Summer.
2019 NSERC USRA, McGill University,
Fellowship awarded by the Canadian government to undergraduate students to pursue research for a Summer.

2017 Alma Mater entrance scholarship, McGill University
Merit-based entrance scholarship to first-time university students entering a full-time undergraduate degree program.

Teaching

W 2026	MATH 1210	*Calculus 1*	Utah
S 2025	MATH 1210	*Calculus 1* (Online)	Utah
W 2025	MATH 1030	*Quantitative reasoning*	Utah
W 2024	MATH 1220	*Calculus II* (Online)	Utah
W 2023	MATH 1320	*Calculus II for engineers* (TA)	Utah

Service

2026-2027 AWM (Association for Women in Mathematics) speaker series cochair
2025-2026 AWM outreach cochair
2024-2025 AWM mentor *Mentoring multiple mathematics undergraduates.*
2024 BRIDGES co-organizer *This is an AWM conference aimed towards early graduate students and advanced undergraduates*
2023-2024 Student commutative algebra seminar organizer
2023-2024 Graduate Student Advisory Committee EDI representative
2021-2022 Polyquity committee member *Polyquity is the association that promotes gender equality on the EPFL campus*
2020-2022 EPFL mentor *Mentoring a group of first year students to help them navigate the EPFL and math.*
2019-2020 Vice-President SUMM. *This is the largest undergraduate math conference in Canada with over a hundred participants annually.*

Talks

Winter 2027 *TBA*, Recent developments in commutative algebra, JMM
Fall 2026 *TBA*, University of Michigan, Commutative Algebra seminar
Fall 2026 *TBA*, Arizona State University AMS Sectional
Summer 2026 *Perfectoid pure pairs*, Université Paris Saclay, Informal Seminar
Winter 2026 *F-singularities of foliations*, North Dakota State University AMS Sectional
Winter 2026 *Centers of perfectoid purity*, Special session on commutative algebra, JMM
Fall 2025 *Centros de pureza perfectoide* (in Spanish), Seminario Chichimeca de Álgebra, UA-Aguascalientes
Fall 2025 *Centers of perfectoid purity*, Algebra seminar, University of Missouri
Fall 2025 *Centers of perfectoid purity*, Workshop on Notions of Singularity in Different Characteristics, Banff International Research Station
Fall 2025 *Centers of perfectoid purity*, Fall Southeastern AMS sectional, Tulane University
Fall 2024 *Centers of perfectoid purity*, CIMAT Algebra seminar
Fall 2024 *Centers of perfectoid purity*, Congreso nacional de la sociedad matemática mexicana
Summer 2024 *Mixed characteristic compatible ideals*, Upcoming Researchers in Commutative Algebra, University of Nebraska Lincoln
Spring 2024 *Tame ramification and centers of F-purity*, CIMAT Commutative Algebra Learning Seminar, CIMAT
Fall 2023 *Tame ramification and centers of F-purity*, Commutative Algebra Seminar, University of Utah
Fall 2023 *On tame ramification and compatible ideals*, BIKES, the Graduate Student Seminar at the University of Utah
Fall 2023 *Compatible ideals and their behavior under finite covers*, Commutative Algebra Reading Seminar, University of Nebraska

Spring 2023 *An introduction to compatible ideals*, Commutative Algebra Graduate Student Seminar at U Michigan

Spring 2023 *An introduction to compatible ideals*, BIKES, the Graduate Student Seminar at the University of Utah

Posters

Centers of perfectoid purity, KUMUNU, University of Missouri, Fall 2025

Centers of perfectoid purity, SRI Bootcamp, University of Colorado Fort Collins, Summer 2025

Centers of perfectoid purity, Singularities in Algebra and Geometry, CIMAT, Summer 2025

Tame ramification and centers of F-purity, Western Algebraic Geometry Symposium, Washington University in St. Louis, Fall 2023.